

## 作業 2.

1. (1) Gauss 定理:  $\oint_V \nabla \cdot \vec{B} dV = \oint_S \vec{B} \cdot d\vec{s}$

取  $\vec{B} = \vec{f} \times \vec{c}$ , 其中  $\vec{c}$  是任意常向量

$$\begin{aligned} \text{左式} &= \oint_V \nabla \cdot (\vec{f} \times \vec{c}) dV = \oint_V [\vec{c} \cdot (\nabla \times \vec{f}) - \vec{f} \cdot (\nabla \times \vec{c})] dV = \oint_V \vec{c} \cdot (\nabla \times \vec{f}) dV \\ &= \vec{c} \cdot \left( \oint_V dV \nabla \times \vec{f} \right) \end{aligned}$$

$$\text{右式} = \oint_S (\vec{f} \times \vec{c}) \cdot d\vec{s} = \oint_S \vec{c} \cdot (d\vec{s} \times \vec{f}) = \vec{c} \cdot \oint_S d\vec{s} \times \vec{f}$$

因此有  $\vec{c} \cdot \left[ \oint_V dV \nabla \times \vec{f} - \oint_S d\vec{s} \times \vec{f} \right] = 0$ , 再由  $\vec{c}$  的任意性知

$$\oint_V dV \nabla \times \vec{f} = \oint_S d\vec{s} \times \vec{f}$$

(2) Stokes 定理:  $\oint_S (\nabla \times \vec{B}) \cdot d\vec{s} = \oint_L \vec{B} \cdot d\vec{l}$

取  $\vec{B} = \varphi \vec{c}$ , 其中  $\vec{c}$  是任意常向量

$$\begin{aligned} \text{左式} &= \oint_S [\nabla \times (\varphi \vec{c})] \cdot d\vec{s} = \oint_S [\nabla \varphi \times \vec{c} + \varphi \nabla \times \vec{c}] \cdot d\vec{s} = \oint_S (\nabla \varphi \times \vec{c}) \cdot d\vec{s} \\ &= \oint_S \vec{c} \cdot (d\vec{s} \times \nabla \varphi) = \vec{c} \cdot \oint_S d\vec{s} \times \nabla \varphi \end{aligned}$$

$$\text{右式} = \oint_L \varphi \vec{c} \cdot d\vec{l} = \vec{c} \cdot \oint_L d\vec{l} \varphi$$

所以  $\vec{c} \cdot \left[ \oint_S d\vec{s} \times \nabla \varphi - \oint_L d\vec{l} \varphi \right] = 0$ , 再由  $\vec{c}$  的任意性的

$$\oint_S d\vec{s} \times \nabla \varphi = \oint_L d\vec{l} \varphi$$

2. (a) 直角坐標系:  $r = \sqrt{x^2 + y^2 + z^2}$

$$\nabla \left( \frac{1}{r} \right) = -\frac{1}{r^2} \nabla r = -\frac{1}{r^2} \left( \frac{x}{r} \hat{i} + \frac{y}{r} \hat{j} + \frac{z}{r} \hat{k} \right) = -\frac{\hat{r}}{r^2} = -\frac{\vec{r}}{r^3} = -\frac{x\hat{i} + y\hat{j} + z\hat{k}}{(x^2 + y^2 + z^2)^{3/2}}$$

(b) 柱座標系:  $r = \sqrt{\rho^2 + z^2}$

$$\nabla \left( \frac{1}{r} \right) = -\frac{1}{r^2} \nabla r = -\frac{1}{r^2} \left( \frac{\rho}{r} \hat{\rho} + 0 \hat{\theta} + \frac{z}{r} \hat{z} \right) = -\frac{\rho \hat{\rho} + z \hat{z}}{r^3} = -\frac{\rho \hat{\rho} + z \hat{z}}{(\rho^2 + z^2)^{3/2}}$$

(c)  $\nabla \left( \frac{1}{r} \right) = -\frac{1}{r^2} \nabla r = -\frac{1}{r^2} (1 \hat{r} + 0 \hat{\theta} + 0 \hat{\phi}) = -\frac{\hat{r}}{r^2}$

3. 圓柱坐標系中,  $\nabla \varphi = \frac{\partial \varphi}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial \varphi}{\partial \theta} \hat{\theta} + \frac{\partial \varphi}{\partial z} \hat{z}$

$$\nabla \times \nabla \varphi = \frac{1}{\rho} \begin{vmatrix} \hat{\rho} & \rho \hat{\theta} & \hat{z} \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ \frac{\partial \varphi}{\partial \rho} & \frac{\partial \varphi}{\partial \theta} & \frac{\partial \varphi}{\partial z} \end{vmatrix} = \frac{1}{\rho} \left( \frac{\partial^2 \varphi}{\partial \theta \partial z} - \frac{\partial^2 \varphi}{\partial z \partial \theta} \right) \hat{\rho} + \left( \frac{\partial^2 \varphi}{\partial z \partial \rho} - \frac{\partial^2 \varphi}{\partial \rho \partial z} \right) \hat{\theta} + \frac{1}{\rho} \left( \frac{\partial^2 \varphi}{\partial \rho \partial \theta} - \frac{\partial^2 \varphi}{\partial \theta \partial \rho} \right) \hat{z} = 0$$

在球坐標系中,  $\nabla \times \vec{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{r} & r \hat{\theta} & r \sin \theta \hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix} = \begin{vmatrix} \frac{1}{r^2 \sin \theta} \hat{r} & \frac{1}{r \sin \theta} \hat{\theta} & \frac{1}{r} \hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$

$$\nabla \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (A_\phi)$$

$$\nabla \cdot (\nabla \times \vec{A}) = \begin{vmatrix} \frac{1}{r^2} \frac{\partial}{\partial r} \frac{r^2}{r \sin \theta} & \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \frac{\sin \theta}{r \sin \theta} & \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \frac{1}{r} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix} = 0$$

4.  $\vec{E}_1 = \frac{k Q_1}{|\vec{r} - \vec{r}_1|^3} (\vec{r} - \vec{r}_1) = \frac{k Q_1}{5\sqrt{5}} (-2, -1, 0)$ ,  $\vec{E}_2 = \frac{k Q_2}{|\vec{r} - \vec{r}_2|^3} (\vec{r} - \vec{r}_2) = \frac{k Q_2}{10\sqrt{10}} (3, 1, 0)$

(a)  $E_x = (\vec{E}_1 + \vec{E}_2) \cdot \hat{x} = 0 \Leftrightarrow -2 \cdot \frac{Q_1}{5\sqrt{5}} - 3 \cdot \frac{Q_2}{10\sqrt{10}} = 0 \Leftrightarrow 4\sqrt{2} Q_1 + 3 Q_2 = 0 \Leftrightarrow \frac{Q_2}{Q_1} = -\frac{4\sqrt{2}}{3}$

(b)  $E_y = (\vec{E}_1 + \vec{E}_2) \cdot \hat{y} = 0 \Leftrightarrow -\frac{Q_1}{5\sqrt{5}} + \frac{Q_2}{10\sqrt{10}} = 0 \Leftrightarrow \frac{Q_2}{Q_1} = 2\sqrt{2}$

5. 以線圈圓心為原點  $O$ , 原點  $O$  與點電荷  $Q'$  的位置  $P$  的連線為  $z$  軸,  $\vec{OP}$  為正方向

建立柱座標系

$$\vec{OP} = h \hat{z}$$

$$d\vec{F} = k \frac{Q' \lambda b d\theta}{(h^2 + b^2)^{3/2}} (b \hat{\rho} + h \hat{z})$$

$$\vec{F} = \oint_L d\vec{F} = k \frac{Q' \lambda b}{(h^2 + b^2)^{3/2}} [b \oint_L \hat{\rho} d\theta + h \oint_L \hat{z} d\theta] = k \frac{Q' \lambda b h}{(h^2 + b^2)^{3/2}} \cdot 2\pi \hat{z} = \frac{Q' \lambda b h}{2\epsilon_0 (h^2 + b^2)^{3/2}} \hat{z} = \frac{Q' Q h}{4\pi \epsilon_0 (h^2 + b^2)^{3/2}} \hat{z}$$

當  $h=0$  時,  $\vec{F}(h=0) = 0$

$$\text{當 } h \gg b \text{ 時, } (h^2 + b^2)^{-3/2} \approx h^{-3} \left[ 1 - \frac{3}{2} \left( \frac{b}{h} \right)^2 \right], \vec{F}(h \gg b) = \frac{Q' Q h}{4\pi \epsilon_0} \cdot h^{-3} \left[ 1 - \frac{3}{2} \left( \frac{b}{h} \right)^2 \right] = \frac{Q' Q}{4\pi \epsilon_0 h^2} \left[ 1 - \frac{3}{2} \left( \frac{b}{h} \right)^2 \right] \approx \frac{Q' Q}{4\pi \epsilon_0 h^2}$$

↑  
二階近似

↑  
一階(零階)近似